Sharp Cubic Smoothing and Law-Preserving Certificates for the ℓ_1 Markov-Cotype Construction

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Abstract

We study the finite-cut and stopped-walk mechanism in the manuscript *Metric Markov Cotype Two of* ℓ_1. For its cubic $\phi(s) = 3s^2 - 2s^3$ and binary-centered quartic potential, we prove that the dimension-independent increment constant is exactly 18, improving the source's valid constant 108. The proof retains the relation between the two endpoint norms and their inner product; a two-coordinate family proves sharpness for this particular inequality. Keeping the initial martingale potential strengthens the approximation coefficient from one to three. Exact factorization of a geometric time into a block number and a remainder improves the source's endpoint comparison, without changing the stopping distribution. These estimates give a numerical refinement of the same finite-chain construction, not a new proof of the external Lipschitz extension theorem or an optimal Markov-cotype constant. We also give cube-preserving capped-time approximations, physical residual bounds, exact examples, and negative controls for nonbinary centers and altered stopping information. Contribution assessment: C3/5, a sharp source-specific flat-map constant and retained-potential and stopping-law refinements give notable quantitative improvements with explicit error and scope boundaries.

1 Model, quantifiers and attribution

All scalars are real unless stated otherwise. Let $A = (a_{ij})$ be an n by n stochastic matrix, π a probability row vector, and $\pi A = \pi$. Zero entries of π are allowed. Reversibility means $\pi_i a_{ij} = \pi_j a_{ji}$; it implies stationarity, but is not used in the estimates below. Let $t \geq 1$ be an integer and let $x_1, \ldots, x_n \in \ell_1(\mathbb{R})$. Define

$$W_t = \sum_{i,j} \pi_i \left(\frac{1}{t} \sum_{s=1}^{t} A^s \right)_{ij} \|x_i - x_j\|_1^2, \quad \mathcal{C}_t(y) = \sum_i \pi_i \|x_i - y_i\|_1^2 + t \sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 .$$

Metric Markov cotype two asks for $\mathcal{C}_t(y) \leq C^2 W_t$ with one C for every such finite reversible input, and with the new points in the *whole target space*. They are not required to lie in the linear span or a prescribed subspace containing the x_i.

The source [1] claims $N_2(\ell_1) \leq 12\sqrt{21}$, equivalently $C^2 = 3024 = 108 \cdot 28$. Its construction combines exact weighted cuts, a flat cubic, a killed-chain martingale, and a metric endpoint comparison. Its remark that stationarity suffices is already in the source and is not a new extension here. We rederive these interfaces with improved constants and explicit approximation costs. The

global Lipschitz extension consequence still invokes the separate theorem of Mendel and Naor [2]; that theorem's proof is not independently reproduced or certified in this report. Source publication and finite tests alone are not proof-validity or acceptance certificates.

The main earned increment is Theorem 3.1, including its sharpness proof. The retained initial term and the exact block law give additional numerical refinements. Classical cut decomposition, Cauchy–Schwarz, conditional expectation, resolvent identities and norm transport are credited as tools, not as new principles.

2 Exact cuts and the two different norms

For each original coordinate k put $v(k) = \min_i x_i(k)$, and for every nonempty proper subset $B \subset [n]$ put

$$b_B(k) = \left(\min_{i \in B} x_i(k) - \max_{i \notin B} x_i(k) \right)_+ .$$

These vectors are in ℓ_1: $|v(k)| \le \sum_i |x_i(k)|$ and $b_B(k) \le 2 \sum_i |x_i(k)|$. Retain the finite family $\mathcal{B}$ with $w_B = \|b_B\|_1 > 0$ and define

$$z_i(B) = \mathbf{1}_{\{i \in B\}}, \qquad T(u) = v + \sum_{B \in \mathcal{B}} u_B b_B, \qquad \|u\|_{1,w} = \sum_B w_B |u_B|, \quad \|u\|_H^2 = \sum_B w_B u_B^2.$$

At most $2^n - 2$ cuts are retained, even when each x_i has infinitely many coordinates. If the family is empty all data coincide. For a fixed k, ordering its distinct values identifies exactly the nested upper cuts at successive gaps. Telescoping those gaps gives $T(z_i)(k) = x_i(k)$. The nonzero terms in the difference for i, j have a common sign, so summing over k proves

$$T(z_i) = x_i, \qquad \|x_i - x_j\|_1 = \|z_i - z_j\|_{1,w} = \|z_i - z_j\|_H^2 , \qquad \|T(u) - T(u')\|_1 \le \|u - u'\|_{1,w} . \quad (1)$$

The last assertion is the triangle inequality and holds for all real u, u', not just vertices. Positivity of b_B is important to this description. These are the source's reconstruction identities.

The Hilbert squared distance in (1) equals an ℓ_1 distance only on binary differences. It is not an identity of norms on the cube $Q = [0,1]^{\mathcal{B}}$. For example, with one unit-weight cut and $u = s, u' = 0$, it compares s to s^2. This distinction rules out replacing the physical smoothing cost by a convenient Gram norm. Repeated or linearly dependent reconstructed columns do not invalidate (1); invertibility of T is neither asserted nor needed.

3 The optimal constant for the source flat map

Let $\phi(r) = 3r^2 - 2r^3$ on $[0,1]$ and $\Phi(u)_B = \phi(u_B)$. This map fixes binary vertices, $\phi(1-r) = 1 - \phi(r)$, and $0 \le \phi'(r) = 6r(1-r) \le 3/2$. For a binary vertex e, write

$$F_e(u) = \|u - e\|_H^4 , \qquad \mathcal{R}_e(u, u') = F_e(u') - F_e(u) - 4 \|u - e\|_H^2 \langle u - e, u' - u \rangle_H .$$

Theorem 3.1 (Sharp flat-map constant)**.** *For every finite positive weight family, every $u, u' \in Q$, and every $e \in \{0,1\}^{\mathcal{B}}$,*

$$\|\Phi(u') - \Phi(u)\|_{1,w}^2 \le 18 \mathcal{R}_e(u, u'). \quad (2)$$

No constant smaller than 18 works for all these data. This sharpness is for the specified Φ and F_e, not for Markov cotype itself.

Proof. Reflect each coordinate with $e_B = 1$. Set $a_B = |u_B - e_B|$ and $b_B = |u'_B - e_B|$. The reflection is an affine isometry of H with a fixed diagonal sign on differences. It takes e to zero and preserves the absolute increments of Φ. In particular the original Bregman remainder becomes

$$R = B^2 + 3A^2 - 4AC, \qquad A = \|a\|_H^2, \quad B = \|b\|_H^2, \quad C = \langle a, b\rangle_H.$$

For $a_B, b_B \in [0,1]$, integration of $0 \le \phi'(r) \le 6r$ gives $|\phi(b_B) - \phi(a_B)| \le 3|b_B^2 - a_B^2|$. Therefore

$$\|\Phi(b) - \Phi(a)\|_{1,w} \le 3 \sum_B w_B |b_B - a_B|(b_B + a_B).$$

Weighted Cauchy–Schwarz bounds the square of the last sum by

$$\|b - a\|_H^2 \, \|b + a\|_H^2 = (A + B)^2 - 4C^2.$$

The exact identity

$$2R - \big((A + B)^2 - 4C^2\big) = (B - A)^2 + 4(A - C)^2 \ge 0 \tag{3}$$

proves (2). Thus no independent maximization of mutually dependent quantities A, B, C is required.

For sharpness use two unit-weight coordinates, $e = 0$, $u = (\epsilon, \epsilon)$ and $u' = (\epsilon + h, \epsilon - h)$, where $0 < h < \epsilon < 1/2$. Monotonicity of ϕ gives

$$\big\|\Phi(u') - \Phi(u)\big\|_1 = \phi(\epsilon + h) - \phi(\epsilon - h) = 12\epsilon(1 - \epsilon)h - 4h^3, \quad R = 8\epsilon^2 h^2 + 4h^4.$$

For fixed ϵ, the ratio tends to $18(1 - \epsilon)^2$ as $h \downarrow 0$. Then $\epsilon \downarrow 0$ gives 18. A single rational sequence, $h = \epsilon^2$, has the same limit and stays in the cube. This is an analytic limiting argument, not an inference from numerical samples. □

Remark 3.2 (Why the source estimate loses a factor six). The source validly bounds the increment by $6 \|u - e\|_H \|u' - u\|_H + 3 \|u' - u\|_H^2$, and estimates its two terms separately by the remainder. Reflection about the actual binary vertex makes both endpoints nonnegative, permitting the factorization through $b^2 - a^2$ and the joint identity (3). The source constant 108 is not a false bound; it is nonoptimal.

3.1 Endpoint and binary-center gates

The vertex condition is indispensable. In one coordinate set $e = u = 1/2$ and $u' = 1/2 + h$. Then the quartic remainder is h^4, while $|\phi(u') - \phi(u)|^2 = (3h/2 - 2h^3)^2$. Their ratio is unbounded as $h \to 0$. Likewise the identity map in place of ϕ fails at $e = u = 0$: its squared increment is h^2, not $O(h^4)$. These examples identify the needed endpoint flatness, rather than merely testing a positive Gram matrix.

Lemma 3.3 (Initial-potential comparison)**.** *For any $u \in Q$ and binary e,*

$$\|\Phi(u) - e\|_{1,w}^2 \le 9F_e(u). \tag{4}$$

The constant 9 is sharp for this inequality.

Proof. Reflect to $e = 0$ and use $\phi(r) = r^2(3 - 2r) \le 3r^2$ before summing and squaring. With one coordinate, the ratio is $(3 - 2r)^2 \to 9$ as $r \downarrow 0$. □

4 The exact martingale and the retained cost

Put $p = 1/(t+1)$ and $q = t/(t+1)$ and define

$$G = p\sum_{s=0}^{\infty} q^s A^s, \qquad h_i = (Gz)_i, \qquad y_i = T(\Phi(h_i)).$$

The matrix G is stochastic and stationary with respect to π. Its convex series converges, $h_i \in Q$, and

$$h_i = pz_i + q\sum_j a_{ij} h_j. \tag{5}$$

Equivalently $h_i = \mathbb{E}_i z_{X_S}$ for an independent time $\mathbb{P}(S = s) = pq^s$, $s \geq 0$; its mean is exactly t.

The filtration must record killing only when it occurs. Begin with $X_0 \sim \pi$. From an alive index i jump to dead i with probability p or to alive j with probability qa_{ij}. Attach h_i to an alive state, z_i to a dead state, and keep dead states constant. Equation (5) makes the attached process M_m a martingale. It converges almost surely to z_{X_S} after S live jumps and one death jump. The binary center $e = z_{X_0}$ is measurable at time zero; revealing S at that time is *not* the filtration just used.

For any cube-valued martingale and initially measurable binary e, the linear term in (2) has conditional mean zero. Telescoping the potentials and using bounded convergence, then monotone convergence for the nonnegative jump sum, proves

$$\mathbb{E}\sum_{m\geq 0} \|\Phi(M_{m+1}) - \Phi(M_m)\|_{1,w}^2 \leq 18(\mathbb{E}F_e(M_\infty) - \mathbb{E}F_e(M_0)). \tag{6}$$

Boundedness follows from the finite cut family and $F_e \leq (\sum_B w_B)^2$. There is no unproved uniform integrability passage across a changing family of cuts.

Set

$$B_0 = \sum_i \pi_i \|z_i - \Phi(h_i)\|_{1,w}^2, \quad E_0 = \sum_{i,j} \pi_i a_{ij} \|\Phi(h_i) - \Phi(h_j)\|_{1,w}^2, \quad D_G = \sum_{i,j} \pi_i G_{ij} \|x_i - x_j\|_1^2.$$

At time m, the expected death-jump cost is $pq^m B_0$ and the live-jump cost is $q^{m+1}E_0$, by stationarity and independence of the survival trials. Their sum over m is exactly $B_0 + tE_0$, including when A is periodic or has several stationary classes. The terminal potential equals D_G by (1).

Theorem 4.1 (Retained approximation cost)**.** *The source's actual points y_i satisfy*

$$3B_0 + tE_0 \leq 18D_G, \qquad 3\sum_i \pi_i \|x_i - y_i\|_1^2 + t\sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \leq 18D_G. \tag{7}$$

Proof. Write $I_0 = \sum_i \pi_i F_{z_i}(h_i)$. Equations (6) and (4) give $B_0 + tE_0 \leq 18(D_G - I_0)$ and $B_0 \leq 9I_0$. Moving $2B_0$ to the left proves the first inequality. Apply the contraction T to both types of increments to obtain the second. □

The coefficient three is a proved consequence of these two estimates, not a claim of the best possible tradeoff for all data. The zero-mass states of π cause no division by zero; they can be retained throughout.

5 Keeping the geometric block law

This section applies to arbitrary metric-valued stationary chains. Write $D(s) = \mathbb{E}d(x_{X_s}, x_{X_0})^2$ and $W = t^{-1}\sum_{s=1}^{t} D(s)$. All quantities are finite since the state space is finite. Stationarity, the metric triangle inequality and Minkowski give

$$\sqrt{D(r+s)} \le \sqrt{D(r)} + \sqrt{D(s)}.$$

In particular $D(t) \le 2D(r) + 2D(t-r)$. Averaging $r = 1, \dots, t$ and keeping the missing $D(t)$ in the second sum gives

$$D(t) \le d_t W, \qquad d_t = \min\left\{t, \frac{4t}{t+2}\right\} \le 4. \tag{8}$$

The bound tW is immediate from nonnegativity.

Theorem 5.1 (Independent-block comparison)**.** *For the nonnegative geometric time with $p = 1/(t+1)$ and $q = 1-p$,*

$$\mathbb{E}D(S) \le \kappa_t W \le (14 + 4\sqrt{2})W, \tag{9}$$

where, writing $v = q^t$,

$$\mu_t = \frac{v}{1-v}, \qquad \nu_t = \frac{v(1+v)}{(1-v)^2}, \qquad \beta_t = \begin{cases} 0, & t = 1, \\ tpq/(1-v), & t \ge 2, \end{cases}$$

$$\kappa_t = \nu_t d_t + 2\mu_t\sqrt{d_t\beta_t} + \beta_t. \tag{10}$$

In particular $\kappa_1 = 3$. These are upper bounds, not optimal comparison constants over all stationary metric data.

Proof. Let $J = \lfloor S/t \rfloor$ and $R = S - tJ$. Direct factorization gives

$$\mathbb{P}(J = j, R = r) = pq^{jt+r} = (1-v)v^j\,\frac{pq^r}{1-v}, \qquad j \ge 0, \quad 0 \le r < t. \tag{11}$$

Thus J, R are independent, $\mathbb{E}J = \mu_t$ and $\mathbb{E}J^2 = \nu_t$. Since $D(0) = 0$, the largest relevant remainder weight for $t \ge 2$ is $pq/(1-v)$, and $\mathbb{E}D(R) \le \beta_t W$. For $t = 1$ it is zero. Repeated subadditivity gives

$$D(Jt + R) \le J^2 D(t) + 2J\sqrt{D(t)D(R)} + D(R).$$

Independence in (11), followed by $\mathbb{E}\sqrt{D(R)} \le \sqrt{\mathbb{E}D(R)}$, proves (10). Finally $(1 + 1/t)^t \ge 2$ gives $v \le 1/2$, hence $\mu_t \le 1$, $\nu_t \le 3$, and $\beta_t \le tp/(1-v) \le 2$. Together with $d_t \le 4$ these yield $12 + 4\sqrt{2} + 2 = 14 + 4\sqrt{2}$. □

The source obtains 28 by squaring the block estimate with a factor two and using a coarser estimate of $\mathbb{E}J^2$. Applying Minkowski before squaring already gives $(2\sqrt{3} + \sqrt{2})^2 = 14 + 4\sqrt{6}$. The smaller constant here additionally uses the exact independence of J and R: the mixed term contains $\mathbb{E}J$, not $\sqrt{\mathbb{E}J^2}$. Arbitrary independent stopping distributions need not give independent quotient and remainder. One cannot reuse this constant after changing the time law merely because its mean is t.

Corollary 5.2 (Numerical consequence for the source construction)**.** *For every finite stationary input in Section 1, the source points $y_i = T(\Phi(Gz)_i)$ obey*

$$3\sum_i \pi_i \|x_i - y_i\|_1^2 + t\sum_{i,j} \pi_i a_{ij} \|y_i - y_j\|_1^2 \le 18\kappa_t W_t \le (252 + 72\sqrt{2})W_t. \tag{12}$$

Consequently the same proof construction yields $N_2(\ell_1(\mathbb{R})) \le \sqrt{252 + 72\sqrt{2}}$.

Proof. Apply Theorem 5.1 to the original ℓ_1 distances in Theorem 4.1. Discarding the extra nonnegative approximation term gives the definition of metric Markov cotype. The all-coincident case uses $y_i = x_i$. The argument has no dimension-dependent limit or finite-fixture premise. □

The numerical conclusion refines the source's own construction; it is not credited here as a new qualitative resolution of the problem. If one elects to retain its original comparison lemma unchanged, Theorem 3.1 alone gives $C^2 = 18 \cdot 28 = 504$. The sharpness of 18 does not show that any of these cotype constants is optimal: extremizing local increments need not be simultaneously realizable along the relevant stationary martingale.

6 Certified finite stopping and inexact solves

Let $V_w = (\mathbb{R}^{\mathcal{B}}, \|\cdot\|_{1,w})$ and define

$$\|f\|_{2,\pi;1,w}^2 = \sum_i \pi_i \|f_i\|_{1,w}^2 .$$

On positive-mass states this is a norm; ignoring zero-mass states gives the same formulas. For any V_w-valued array, triangle inequality, scalar Jensen and stationarity show $\|Af\|_{2,\pi;1,w} \le \|f\|_{2,\pi;1,w}$. No Euclidean or spectral-radius substitution is made.

Proposition 6.1 (Physical residual certificate)**.** *Take any candidate $\widehat{h}_i \in Q$ and form its* actual *residual $r = pz + qA\widehat{h} - \widehat{h}$. Let $\widehat{y}_i = T(\Phi(\widehat{h}_i))$ and $\rho = \|r\|_{2,\pi;1,w}$. Then*

$$\left\|h - \widehat{h}\right\|_{2,\pi;1,w} \le (t+1)\rho, \tag{13}$$

$$\sqrt{\mathcal{C}_t(\widehat{y})} \le \sqrt{18\kappa_t W_t} + \frac{3}{2}(t+1)\sqrt{1+4t}\,\rho. \tag{14}$$

Proof. Subtract the equations to get $(I - qA)(h - \widehat{h}) = r$. The induced-norm Neumann series converges with norm at most $1/(1-q) = t+1$, proving the first bound. Since $\sup_{[0,1]} |\phi'| = 3/2$, contraction of T gives $\|\widehat{y} - y\|_{2,\pi;\ell_1} \le (3/2)(t+1)\rho$.

For any error array e_i, stationarity yields

$$\sum_{i,j} \pi_i a_{ij} \|e_i - e_j\|_1^2 \le 2\sum_{i,j} \pi_i a_{ij}(\|e_i\|_1^2 + \|e_j\|_1^2) = 4\sum_i \pi_i \|e_i\|_1^2 .$$

Apply Minkowski in the direct sum of the state and edge L_2 spaces to the two components of $\mathcal{C}_t$; the error operator has norm at most $\sqrt{1+4t}$. Corollary 5.2 bounds the exact construction, completing the proof. □

The cube gate is real: outside $[0,1]$ the stated derivative bound is false. If a candidate is clipped to the cube, its residual must be recomputed *after* clipping. An omitted component of the residual is not certified by a small projected residual. This proposition concerns all cut coordinates and all positive-mass states. A norm estimate on a few sampled entries is not its premise.

Proposition 6.2 (Capped time preserves a probability law)**.** *For an integer $K \geq 0$ set*

$$G_K = p \sum_{s=0}^{K-1} q^s A^s + q^K A^K, \qquad h^{(K)} = G_K z, \tag{15}$$

with an empty sum when $K = 0$. It is the law of $X_{S \wedge K}$, is stochastic, satisfies $\pi G_K = \pi$, and maps the binary data into Q. The exact identities

$$h - h^{(K)} = q^K A^K (h - z), \tag{16}$$

$$pz + qAh^{(K)} - h^{(K)} = q^{K+1} A^K (A - I) z \tag{17}$$

hold without reversibility. If $D_* = \max_{i,j} \|x_i - x_j\|_1$, then

$$\sqrt{\mathcal{C}_t(T\Phi(h^{(K)}))} \leq \sqrt{18\kappa_t W_t} + \frac{3}{2}\sqrt{1+4t}\, q^K D_*. \tag{18}$$

Proof. The weights in (15) sum to 1; the last weight is the tail probability $\mathbb{P}(S \geq K)$. The full geometric tail is $q^K A^K G$, which proves (16); multiplying out the finite sum proves (17). Each $\|h_i - z_i\|_{1,w} \leq \sum_j G_{ij} \|z_j - z_i\|_{1,w} \leq D_*$ by (1). A stochastic matrix is contractive in the maximum of these norms, so (16) bounds the uniform error by $q^K D_*$. Use the same cubic and state/edge bounds as above. □

For fixed t, the error decays with K. As t grows, $q = t/(t+1)$ approaches one; the required stopping depth need not stay bounded. Computing or storing all cuts can cost exponentially in n. These are certificates, not measured speedups or a dimension-free finite algorithm for Lipschitz extension.

Dropping the final term of (15) gives row sum $1 - q^K$, not one. Such a partial inverse is algebraically useful but is not an endpoint probability law. Even its constant mode is damped. Renormalizing the partial sum gives a different distribution, not the original geometric law or the capped law, and invalidates the exact residual formulas unless rederived.

7 Worked finite objects and negative controls

7.1 Two-state reversible walk

Take $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\pi = (1/2, 1/2)$, $t = 1$, and scalar data $x_0 = 0, x_1 = 1$. The single cut has unit weight and

$$G = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad h = (1/3, 2/3), \quad y = (7/27, 20/27).$$

Here $D_G = 1/3$, $W_1 = 1$, $B_0 = 49/729$, $E_0 = 169/729$, and $3B_0 + E_0 = 316/729 \leq 18 D_G = 6$. The inequality need not nearly saturate its local sharp constant. At $K = 1$ the capped kernel is $(I + A)/2$, giving $\widehat{h} = (1/2, 1/2)$; at $K = 0$ it is I. Thus the approximations stay in the cube and have transparent probability meanings.

For this example, revealing $S = 0$ initially would make the next attached value deterministically z_i, whereas the starting value is $h_i \neq z_i$. Its conditional expectation would not equal the starting value. This explicitly explains why the source filtration reveals successive survival trials rather than the entire stopping time in advance.

7.2 Ambient reconstruction can leave a subspace

Use three vectors $x_1 = (1, 0, 1)$, $x_2 = (0, 1, 1)$, $x_3 = (0, 0, 0)$ in the plane $L = \{x : x_3 = x_1 + x_2\}$. Let $A_{ij} = 1/3$, π uniform and $t = 1$. Then $G = (I + A)/2$ has diagonal $2/3$ and off-diagonal $1/6$. The median of one copy of each vector is $(0, 0, 1) \notin L$; if an index is repeated, the median is one of the original vectors. Three independent endpoints with row law $G_{i\cdot}$ therefore give

$$y_i(3) - y_i(1) - y_i(2) = 6G_{i1}G_{i2}G_{i3} = 1/9.$$

The source's median interpretation follows from threshold cuts: the chance of at least two of three successes with probability h is $3h^2 - 2h^3$. Hence these are precisely its points, not alternative averages. This example shows why a subspace restriction cannot be silently added to metric Markov cotype.

7.3 Stationarity is not detailed balance

For the directed three-cycle A, with uniform π and $t = 1$, $G = (4I + 2A + A^2)/7$. It is stochastic and stationary though A is not reversible. All stopped-cost and comparison identities above still apply. Conversely, if the starting law is not stationary, the live-state distribution is πA^m, not π; the jump sum cannot simply be replaced by $B_0 + tE_0$. No extension to arbitrary initial laws with unchanged weights is claimed.

8 Ram representations with the actual norms

The original private physical-metric chapter [6], read in full at its stated scope, proves that synthesis J carries the metric J^*J, and separates projected from full residuals. Its original proof does not say that every norm is quadratic. For an invertible finite change of cut coordinates $u = Uc$, the faithful description here is

$$\|u\|_H^2 = c^{\mathsf{T}}U^{\mathsf{T}}\operatorname{diag}(w)Uc, \qquad \|u\|_{1,w} = \sum_B w_B|(Uc)_B|, \qquad 0 \le Uc \le 1.$$

Moreover the cubic becomes $c \mapsto U^{-1}\Phi(Uc)$, not the coordinatewise cubic in c. The binary vertices become $U^{-1}\{0,1\}^{\mathcal{B}}$, not arbitrary binary vectors in the new coordinates. A polynomial identity or positive Gram matrix that omits any of these transported objects proves a different statement.

For instance with unit weights and $U = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the actual cut norm is $|c_1 + c_2| + |c_2|$ and the Hilbert square is $(c_1 + c_2)^2 + c_2^2$. The former is not a Hilbert norm: already ℓ_1^2 violates the parallelogram law at the two coordinate unit vectors. This is not a defect to cure by whitening; both geometries play different roles in the proof.

The original restricted-dual proof in [4] retains the physical metric and the unresolved component. Its coercive residual theorem requires the full dual norm on the declared domain, not a finite list of tests. The original induced-norm Neumann proof in [5] requires a genuine contraction and keeps the order of factors. Proposition 6.1 is a concrete specialization of that classical residual mechanism: qA contracts in the specified mixed norm because of positivity and stationarity. The useful new part is its integration with the flat cubic, reconstructed output and state/edge cost.

The later comparison [7] explains why an approximate polynomial identity needs a residual budget. Here (3) is an exact polynomial identity and its two squares are nonnegative for every admissible input; this avoids fitting an SOS matrix to samples. It still does not permit replacing the positive cut weights, common binary center, or stopped-martingale filtration. None of these original or comparative Ram proofs alone establishes the source's qualitative extension theorem.

9 Source dependencies and current literature

The scientific source is a single 648-line file, with a bibliography and README. The logical map is small enough to state explicitly.

Source lines	Mechanism	Treatment here
49–188	Definition and global claims	Actual ambient target and all finite-chain quantifiers retained.
190–268	Finite cuts, reconstruction	Reproved with weights, coordinate order and contraction.
270–373	Cubic and quartic martingale	Sharp18 replaces valid108; initial potential retained.
375–446	Geometric/Cesaro comparison	Exact independent blocks yield (9).
448–591	Stopped law, assembly, medians	Same filtration, killing jump, stationary law and reconstruction.
593–643	Complex target and extension	Distortion and external extension hypotheses retained; no independent proof of the external theorem.

No mathematical contradiction or correction to a false source statement was found in these inspected steps. The changes concern valid but nonoptimal estimates and quantified approximation. For complex ℓ_1, splitting into real and imaginary coordinates has distortion at most $\sqrt{2}$; the same source argument increases the squared constant by at most two. This is inherited, not a new complex-geometry theorem.

The public primary refresh on October 9, 2026 used only public titles and URLs. The publisher repository hosts the October 5 source [1]; the cached-byte comparison with captured October trees is provenance, not live proof validation. The version record and relevant statements of Mendel–Naor [2] were checked: Theorem 1.11 also requires a W_p-barycentric target, supplied by ordinary expectation in a Banach space with constant one. Corollary 1.13 uses a dual Banach target for global extension; $\ell_1 = c_0^*$ satisfies that premise. Its extension loss is an *implicit universal multiple* of Markov type and cotype constants, not the exact number in Corollary 5.2. The extension proof itself remains an external input.

The primary abstract and version record of Cheng–Wang–Xiang [3], version 2 dated September 9, 2026, concern torus *metric cotype* and nonlinear cut smoothing. That invariant is distinct from metric Markov cotype, and this abstract is not a proof of the source's stopped-chain theorem. No full audit of that external proof or exhaustive historical-priority search is claimed. The source's 2026 qualitative claim should not be described as an established community consensus merely because its PDF is public.

10 Advantages, limits and evidence

The advantage of the sharp increment inequality is a genuinely uniform constant for every cut family, with a complete sharpness argument for the local analytic problem. The approximation bonus keeps information the source discarded. The independent-block comparison uses exactly the same random time and therefore needs no new coupling or law-identification assumption. Capped stopping offers a finite, positive, normalized approximation with an exact residual. Its costs are explicit growth with t, possible exponential cut storage, and potentially conservative state/edge propagation.

There are several honest remaining boundaries. We have not found the optimal Markov-cotype constant, optimized the geometric law against the killing cost, or proved a best finite truncation

depth. The local sharp family does not prove sharpness of the global chain bound. A different flat map would require a new remainder analysis. Omitting cuts requires a separate estimate of their physical mass and their effect on reconstruction. Alternative stopping times require fresh conditional expectations and endpoint comparisons. None of these extensions follows from a finite matrix being invertible or from the test receipt.

10.1 Literal proof scope and independent checks

All 716 lines of the three cached source texts were displayed and read in bounded chunks after their pre-read hashes were persisted. The ledger records each actual interval, post-read hash and unread complement. Private reading was limited to the named original theorem bodies and the physical-metric chapter, together with selected passages of the later comparison. Included figures and external references are not silently counted as read proofs. These scopes are not an audit of the entire confidential corpus.

The owned executable checks passed 308 exact cases and 4,942 assertions, with zero floating-point diagnostic cases. They are rational and symbolic fixtures: weighted cube inequalities, the sharpness limit, actual cut reconstructions, geometric block probabilities, reversible and nonreversible stationary chains, zero stationary mass, capped stopping, and physical residual identities. They are supplemental to the proofs and include negative controls for nonbinary centers, nonflat maps and unnormalized partial kernels. Their case and assertion counts are recorded separately; no parent or foreign cases are included. The final evidence records the compiled source hash, exported PDF hash and individual inspection of every page.

The contribution is therefore a quantitative refinement of the source's specific finite-cut construction and its exact stopping interfaces. It is not a rebranding of a generic Hilbert coordinate calculation and does not claim independent historical ownership of the source's qualitative global conclusion.

References

[1] OpenAI. *Metric Markov Cotype Two of ℓ_1*, October 5, 2026. Pinned manuscript, main file, bibliography and README read in full. Public repository source.

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[4] M. Subbarao. Original local confidential TR-core-pat-4. Restricted dual reconstruction, physical energy gate and finite-test limitations; exact read intervals in the ledger.

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[7] M. Subbarao. Local confidential TR-item-047-astra, later comparative report. Exact atomic support and residual-aware polynomial certificate passages, scoped reading only; not a replacement for original proofs.